Sector decomposition for an anisotropic oscillatory integral

1 Setting and overview

1.1 The integral

Let $\ell_1, \dots, \ell_d \geq 0$ and, for $s > 0$, let $A_s = \operatorname{diag}(s^{\ell_1}, \dots, s^{\ell_d})$. For $t \geq 1$ and $\xi \in \mathbb{R}^d$ we define the phase

$$\psi(t, \xi) = \int_1^t |A_s \xi| \, ds.$$

We study the oscillatory integral

$$I(t, x) = \int_{\mathbb{R}^d} e^{i(x \cdot \xi - \psi(t,\xi))} \, |A_t \xi|^{-1} \, \chi(\xi) \, d\xi,$$

where χ is a smooth cutoff supported in the annulus $\{\frac{1}{2} \leq |\xi| \leq 2\}$. We want to bound $I(t, x)$ for large t, uniformly in x.

1.2 The difficulty and the plan

The matrix A_t stretches each frequency coordinate by a different power of t. So there is no single large parameter in front of the phase. We handle this in two moves.

First we split frequency space into d sectors, according to which weighted coordinate $t^{\ell_j}|\xi_j|$ is the largest. On sector k we use the dominant coordinate ξ_k and the $d-1$ ratios $\eta_j = t^{\ell_j}\xi_j/(t^{\ell_k}\xi_k)$ as new variables. In these variables the sector integral becomes an outer integral in ξ_k of an inner oscillatory integral in η with a single large parameter $\lambda = t^{\ell_k+1}|\xi_k|$ (Section 2).

Second we bound the inner integral by stationary phase. The phase is strictly concave in η, and its curvature in each direction is at least of order 1; in borderline directions it reaches order $\log t$ when the leading coordinate takes over early enough (Section 3). Its higher derivatives are not bounded uniformly in t, but they are once we weight them correctly (Section 4). The stationary phase theorem needs its constants to be bounded uniformly in t. At unit scale in η they are not; at the natural scale of each direction, $(\lambda L_j)^{-1/2}$ with $L_j = 1$ or $L_j = \log t$, they are. Section 5 checks this, and Section 6 assembles the main result, Theorem 5. The outer integral in ξ_k is not treated here.

1.3 Conventions

In all estimates $t \geq 2$. The index k is fixed, $n = d - 1$, and i, j, m run over $\{1, \dots, d\} \setminus \{k\}$. Multi-indices $\alpha = (\alpha_j)_{j \neq k}$ have $d - 1$ entries, and δ_i is the multi-index with a 1 in position i and 0 elsewhere. We write C, c for positive constants that depend only on d and $\ell_1, \dots, \ell_d$ and may change from line to line. C_α may also depend on α.

2 Reduction to one sector

2.1 Sectors

Sector k is the set of ξ for which $t^{\ell_k}|\xi_k|$ is the largest of the weights $t^{\ell_1}|\xi_1|, \dots, t^{\ell_d}|\xi_d|$. Since $|A_t\xi|^2$ is the sum of the d squared weights, on sector k

$$t^{\ell_k}|\xi_k| \;\geq\; \frac{|A_t\xi|}{\sqrt{d}}.$$

Every ξ lies in some sector, so I is a sum of d sector integrals. We fix k.

2.2 New variables

For $j \neq k$ we introduce the ratios

$$\eta_j = \frac{t^{\ell_j}\xi_j}{t^{\ell_k}\xi_k},$$

which compare the weight of coordinate j with the dominant weight. On sector k they satisfy $|\eta_j| \leq 1$. We write $\eta = (\eta_j)_{j\neq k} \in \mathbb{R}^{d-1}$ and use (ξ_k, η) as coordinates on the sector.

2.3 The integrand in the new variables

We rewrite each factor of the integrand in terms of ξ_k and η.

The amplitude. Since

$$|A_t\xi|^2 = t^{2\ell_k}\xi_k^2 + \sum_{j\neq k} t^{2\ell_j}\xi_j^2 = t^{2\ell_k}\xi_k^2\Big(1 + \sum_{j\neq k}\eta_j^2\Big),$$

we have

$$|A_t\xi| = t^{\ell_k}|\xi_k|\,(1+|\eta|^2)^{1/2}, \qquad |A_t\xi|^{-1} = t^{-\ell_k}|\xi_k|^{-1}(1+|\eta|^2)^{-1/2}.$$

The phase ψ. In the same way, for $0 < s \leq t$,

$$|A_s\xi| = s^{\ell_k}|\xi_k|\Big(1 + \sum_{j\neq k}\frac{s^{2\ell_j}\xi_j^2}{s^{2\ell_k}\xi_k^2}\Big)^{1/2}.$$

We measure time in units of t: substitute $s = \sigma t$, so $ds = t\,d\sigma$ and $s^{2(\ell_j-\ell_k)}\xi_j^2/\xi_k^2 = \sigma^{2(\ell_j-\ell_k)}\eta_j^2$. This gives

$$\psi(t,\xi) = t^{\ell_k+1}|\xi_k|\,G(t,\eta), \qquad G(t,\eta) = \int_{1/t}^{1}\sigma^{\ell_k}\Big(1 + \sum_{j\neq k}\sigma^{2(\ell_j-\ell_k)}\eta_j^2\Big)^{1/2}d\sigma.$$

So the phase is a large factor $t^{\ell_k+1}|\xi_k|$ times a function G of η, whose only dependence on t is through the lower limit $1/t$.

The linear term. We aim at a bound uniform in x, so we may replace x by $A_t x$ without changing $\sup_x |I(t,x)|$. The linear part of the phase becomes

$$A_t x\cdot\xi = x_k t^{\ell_k}\xi_k + \sum_{j\neq k} t^{\ell_j}x_j\xi_j = x_k t^{\ell_k}\xi_k + t^{\ell_k}\xi_k\sum_{j\neq k}x_j\eta_j.$$

The measure. For fixed ξ_k, the change of variables $\xi_j = t^{\ell_k-\ell_j}\xi_k\eta_j$, $j \neq k$, gives

$$d\xi = |\xi_k|^{d-1}\prod_{j\neq k} t^{\ell_k-\ell_j}\; d\xi_k\, d\eta = t^{\,d\ell_k-\sum_{j=1}^d \ell_j}\,|\xi_k|^{d-1}\, d\xi_k\, d\eta.$$

2.4 The sector integral

Denote by I_k the part of I coming from sector k. Collecting the formulas above,

$$I_k = t^{\,d\ell_k-\sum_j \ell_j-\ell_k}\int_{\mathbb{R}} e^{ix_k t^{\ell_k}\xi_k}\,|\xi_k|^{d-2}\,J(\xi_k)\,d\xi_k, \qquad J(\xi_k) = \int_{[-1,1]^{d-1}} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta,$$

where the large parameter, the phase and the amplitude are

$$\lambda = t^{\ell_k+1}|\xi_k|, \qquad \Phi(\eta) = \frac{\xi_k}{|\xi_k|}\,\frac{1}{t}\sum_{j\neq k} x_j\eta_j - G(t,\eta),$$

$$b(\xi_k,\eta) = (1+|\eta|^2)^{-1/2}\,\chi\Big(\xi_k e_k + \sum_{j\neq k} t^{\ell_k-\ell_j}\xi_k\eta_j\, e_j\Big).$$

The rest of these notes bounds the inner integral J. Two remarks on its ingredients.

The phase. Φ depends on x and on the sign of ξ_k only through its linear part. All derivatives of order two and higher are those of $-G$ and do not see x at all.

The amplitude and the size of λ. The second factor of b is the cutoff χ in the new variables. Since $|\xi| = |\xi_k|\big(1+\sum_{j\neq k} t^{2(\ell_k-\ell_j)}\eta_j^2\big)^{1/2}$, the amplitude b is supported where

$$\frac{1}{2} \le |\xi_k|\Big(1+\sum_{j\neq k} t^{2(\ell_k-\ell_j)}\eta_j^2\Big)^{1/2} \le 2.$$

With $\ell_{\min} = \min_j \ell_j$ and $|\eta_j| \le 1$, this gives

$$\frac{t^{\ell_{\min}-\ell_k}}{2\sqrt{d}} \;\le\; |\xi_k| \;\le\; 2 \qquad \text{on the support of } b,$$

and therefore $\lambda \ge t^{1+\ell_{\min}}/(2\sqrt{d})$. So λ is indeed large. Note, however, that a derivative ∂_{η_j} falling on the cutoff produces a factor $t^{\ell_k-\ell_j}\xi_k$, which is large when $\ell_j < \ell_k$. This will be the main obstacle in Section 5.

2.5 Types of directions

How each direction $j \neq k$ is treated depends on the number

$$\kappa_j = 2\ell_j - \ell_k + 1.$$

We call direction j

- *good* if $\ell_j \ge \ell_k$, and *bad* if $\ell_j < \ell_k$;
- *oscillatory* if $\kappa_j > 0$, *borderline* if $\kappa_j = 0$, and *flat* if $\kappa_j < 0$.

We write $\mathcal{O}$, $\mathcal{Z}$ and $\mathcal{F}$ for the sets of oscillatory, borderline and flat directions, $\mathcal{T} = \mathcal{Z} \cup \mathcal{F}$ for the non-oscillatory ones, and $\mathcal{B}$ for the bad oscillatory ones. Every good direction is oscillatory, since then $\kappa_j \geq \ell_k + 1 > 0$. So borderline and flat directions are bad. If there is a bad direction we put

$$\theta = \tfrac{1}{2} \min_{m \text{ bad}} (\ell_k - \ell_m) > 0.$$

In short: oscillatory directions gain from oscillation, flat directions gain from the small size of their support, and borderline directions gain from oscillation with an extra factor $(\log t)^{-1/2}$ when the curvature has time to build up.

3 The Hessian of the phase

Set

$$H(\sigma, \eta) = \Big(1 + \sum_{j \neq k} \sigma^{2(\ell_j - \ell_k)} \eta_j^2\Big)^{1/2}, \qquad \text{so that} \qquad G(t, \eta) = \int_{1/t}^{1} \sigma^{\ell_k}\, H(\sigma, \eta)\, d\sigma.$$

The goal of this section is to show that Φ is strictly concave in η, and to measure its curvature in each direction.

First derivatives. Since $\partial_{\eta_j} H = \sigma^{2(\ell_j - \ell_k)} \eta_j\, H^{-1}$,

$$\partial_{\eta_j} \Phi(\eta) = \frac{\xi_k}{|\xi_k|}\, \frac{x_j}{t} - \eta_j \int_{1/t}^{1} \sigma^{2\ell_j - \ell_k}\, H^{-1}\, d\sigma.$$

Second derivatives. The linear part of Φ has zero Hessian, so $\nabla_\eta^2 \Phi = -\nabla_\eta^2 G$. Differentiating once more,

$$\partial_{\eta_i} \partial_{\eta_j} G(t, \eta) = \int_{1/t}^{1} \sigma^{\ell_k}\, H^{-3} \Big(\delta_{ij}\, \sigma^{2(\ell_j - \ell_k)}\, H^2 - \sigma^{2(\ell_i + \ell_j - 2\ell_k)} \eta_i \eta_j \Big)\, d\sigma.$$

The quadratic form. Let $D(\sigma) = \operatorname{diag}\big(\sigma^{2(\ell_j - \ell_k)}\big)_{j \neq k}$. In matrix form the second derivatives read

$$-\nabla_\eta^2 \Phi = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3} \Big(H^2 D - (D\eta)(D\eta)^T \Big)\, d\sigma.$$

Fix $v \in \mathbb{R}^{d-1}$ and put $z = D^{1/2}\eta$, $u = D^{1/2} v$. Then $\langle Dv, v \rangle = |u|^2$, $\langle D\eta, v \rangle = \langle z, u \rangle$, and $|z|^2 = \langle D\eta, \eta \rangle = H^2 - 1$. Hence

$$-\langle \nabla_\eta^2 \Phi\, v, v \rangle = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3} \Big(H^2 |u|^2 - \langle z, u \rangle^2 \Big)\, d\sigma.$$

By Cauchy–Schwarz, $\langle z, u \rangle^2 \leq |z|^2 |u|^2 = (H^2 - 1)|u|^2$, so the bracket lies between $|u|^2$ and $H^2 |u|^2$. Writing

$$S(\sigma, v) = |u|^2 = \sum_{j \neq k} \sigma^{2(\ell_j - \ell_k)} v_j^2,$$

we arrive at the two-sided bound

$$\int_{1/t}^{1} \sigma^{\ell_k} H^{-3}\, S\, d\sigma \;\leq\; -\langle \nabla_\eta^2 \Phi\, v, v \rangle \;\leq\; \int_{1/t}^{1} \sigma^{\ell_k} H^{-1}\, S\, d\sigma.$$

Strict concavity. For $v \neq 0$ the left side is positive. So $-\nabla_\eta^2 \Phi$ is positive definite and Φ is strictly concave in η, for every t, x and ξ_k.

The curvature in each direction. The lower bound is diagonal. Since $\sigma^{\ell_k}\sigma^{2(\ell_j-\ell_k)} = \sigma^{2\ell_j-\ell_k} = \sigma^{\kappa_j-1}$, it reads

$$-\langle \nabla_\eta^2 \Phi\, v, v\rangle \;\geq\; \sum_{j\neq k} \Lambda_j(\eta)\, v_j^2, \qquad \Lambda_j(\eta) = \int_{1/t}^1 \sigma^{\kappa_j-1}\, H(\sigma,\eta)^{-3}\, d\sigma.$$

On a dyadic time interval $\sigma \in [2^{-i-1}, 2^{-i}]$ where H stays bounded, the integrand of Λ_j contributes about $2^{-i\kappa_j}$. So the curvature of an oscillatory direction is built at late times, that of a borderline direction equally at every time scale, and that of a flat direction mostly at early times. To bound Λ_j from below we need to know when H is bounded.

The switch time. For a bad direction m, the term $\sigma^{2(\ell_m-\ell_k)}\eta_m^2$ of H^2 is at most 1 exactly when $\sigma \geq |\eta_m|^{1/(\ell_k-\ell_m)}$. We call

$$\sigma_*(\eta) = \max_{m \text{ bad}} |\eta_m|^{1/(\ell_k-\ell_m)}$$

the switch time, with $\sigma_* = 0$ if there is no bad direction. In the original variables, $\sigma^{\ell_m-\ell_k}|\eta_m| = s^{\ell_m}|\xi_m|/(s^{\ell_k}|\xi_k|)$ with $s = \sigma t$. So $s_* = \sigma_* t$ is the time at which the weight $s^{\ell_k}|\xi_k|$ overtakes every bad weight $s^{\ell_m}|\xi_m|$.

Lemma 1 (Curvature)**.** *Let $R \geq 1$, $t \geq 2$ and $|\eta| \leq R$. Then for all $v \in \mathbb{R}^{d-1}$,*

$$-\langle \nabla_\eta^2 \Phi(\eta)\, v, v\rangle \;\geq\; c_R \sum_{j\neq k} \Gamma_j(\eta)\, v_j^2, \qquad \Gamma_j(\eta) = \int_{\sigma_0}^1 \sigma^{\kappa_j-1}\, d\sigma,$$

where $\sigma_0 = \max\big(1/t, \min(\sigma_*(\eta), \frac{1}{2})\big)$ and $c_R > 0$ depends only on R, d and the ℓ_j. In particular:

(i) $\Gamma_j \geq c$ for every j, so $-\nabla_\eta^2 \Phi \geq c_R\, I$ on $|\eta| \leq R$;

(ii) if j is borderline and $\sigma_*(\eta) \leq t^{-1/2}$, then $\Gamma_j(\eta) \geq \frac{1}{2}\log t$.

Proof. We show that $H \leq C_R$ on $[\sigma_0, 1]$. A good m contributes $\sigma^{2(\ell_m-\ell_k)}\eta_m^2 \leq R^2$. A bad m contributes at most 1 when $\sigma \geq \sigma_*$, and at most $4^{\ell_k-\ell_m}R^2$ when $\sigma \geq \frac{1}{2}$; one of the two holds on $[\sigma_0, 1]$. So $H^{-3} \geq c_R$ there, and since the integrand of Λ_j is nonnegative, $\Lambda_j \geq c_R\Gamma_j$. For (i), $\sigma_0 \leq \frac{1}{2}$, so $\Gamma_j \geq \int_{1/2}^1 \sigma^{\kappa_j-1}d\sigma$, a positive constant. For (ii), $\kappa_j = 0$ gives $\Gamma_j = \log(1/\sigma_0)$, and $\sigma_0 \leq \max(1/t, t^{-1/2}) = t^{-1/2}$. □

Part (ii) is where $\log t$ enters. A borderline direction collects the same amount $\log 2$ of curvature from every dyadic time scale after the switch. If the switch comes before $t^{1/2}$ in physical time, about $\frac{1}{2}\log_2 t$ scales contribute. For flat directions Γ_j is even larger, but there the amplitude is too rough at the stationary phase scale, and this curvature is not used.

An upper bound. Conversely, the upper half of the two-sided bound and $H \geq 1$ give

$$\partial_{\eta_j}^2 G \;\leq\; \bar\Lambda_j = \int_{1/t}^1 \sigma^{\kappa_j-1}\, d\sigma,$$

with $\bar\Lambda_j \leq 1/\kappa_j$ for oscillatory j and $\bar\Lambda_j = \log t$ for borderline j. So after an early switch, the curvature of a borderline direction is exactly of order $\log t$.

4 Higher derivatives of the phase

Let $|\alpha| \geq 2$. Two derivatives kill the linear part of Φ, so $\partial_\eta^\alpha \Phi = -\partial_\eta^\alpha G$, and it suffices to bound $\partial_\eta^\alpha G$. We first give a direct bound. Then we explain why it is not uniform in t, and we prove a weighted bound that is.

4.1 A direct bound

Reduction to one function. For fixed σ put $y_j = \sigma^{\ell_j - \ell_k}\eta_j$, $j \neq k$, and $\langle y\rangle = (1+|y|^2)^{1/2}$. Then $H(\sigma,\eta) = \langle y\rangle$. Since each y_j is a multiple of η_j alone, the chain rule gives

$$\partial_\eta^\alpha H = \prod_{j\neq k} \sigma^{\alpha_j(\ell_j-\ell_k)} \left(\partial_y^\alpha \langle y\rangle\right).$$

Derivatives of $\langle y\rangle$. We claim that for $|\alpha| \geq 1$, $\partial_y^\alpha \langle y\rangle$ is a finite linear combination of terms

$$y^\beta \langle y\rangle^{1-2m} \qquad \text{with} \qquad m \geq 1, \quad 2m - |\beta| = |\alpha|.$$

For $|\alpha| = 1$ this holds, since $\partial_{y_i}\langle y\rangle = y_i\langle y\rangle^{-1}$. Differentiating $y^\beta\langle y\rangle^{1-2m}$ in y_i gives two terms: $\beta_i\, y^{\beta-\delta_i}\langle y\rangle^{1-2m}$, which vanishes if $\beta_i = 0$, and $(1-2m)\, y^{\beta+\delta_i}\langle y\rangle^{1-2(m+1)}$. In both terms $2m-|\beta|$ increases by one, which proves the claim. Since $|y^\beta| \leq \langle y\rangle^{|\beta|}$, each term is bounded by $\langle y\rangle^{|\beta|+1-2m} = \langle y\rangle^{1-|\alpha|}$, so

$$\left|\partial_y^\alpha \langle y\rangle\right| \leq C_\alpha \langle y\rangle^{1-|\alpha|}.$$

In words, $\langle y\rangle$ behaves like a function homogeneous of degree one.

The bound. Differentiating under the integral sign and using $\langle y\rangle = H \geq 1$, we obtain for $|\alpha| \geq 1$

$$\left|\partial_\eta^\alpha G(t,\eta)\right| \leq C_\alpha \int_{1/t}^1 \sigma^{\,\ell_k + \sum_{j\neq k}\alpha_j(\ell_j-\ell_k)}\, H(\sigma,\eta)^{1-|\alpha|}\, d\sigma \leq C_\alpha \int_{1/t}^1 \sigma^{\,\ell_k + \sum_{j\neq k}\alpha_j(\ell_j-\ell_k)}\, d\sigma.$$

The last integral stays bounded as $t\to\infty$ only if the exponent of σ is larger than -1. When some $\ell_j < \ell_k$, this fails for large α_j. We look at this more closely.

4.2 Bad directions and weights

Recall from Section 2.5 that direction j is bad if $\ell_j < \ell_k$.

Why the direct bound fails. High derivatives in a bad direction blow up at $\eta_j = 0$. Indeed, set all η_m with $m \neq j$ to zero and expand $H = (1+\sigma^{2(\ell_j-\ell_k)}\eta_j^2)^{1/2}$ in powers of η_j. The coefficient of η_j^{2N} is $\binom{1/2}{N}\sigma^{2N(\ell_j-\ell_k)}$, so

$$\partial_{\eta_j}^{2N} G(t,0) = (2N)!\binom{1/2}{N}\int_{1/t}^1 \sigma^{\,\ell_k + 2N(\ell_j-\ell_k)}\, d\sigma.$$

Since $\ell_j < \ell_k$, the exponent is below -1 for N large, and the integral tends to infinity as $t\to\infty$. The reason is the switch of Section 3: near $\eta_j = 0$ the coordinate j dominates H at early times $\sigma \sim 1/t$, where $\sigma^{\ell_j-\ell_k}$ is huge. Away from $\eta_j = 0$ this does not happen.

The weights. We measure derivatives in units of the distance to these hyperplanes:

$$v_j(\eta) = 1 \ \ (j \text{ good}), \qquad v_j(\eta) = |\eta_j| \ \ (j \text{ bad}).$$

4.3 The weighted bound

Derivatives of H, weighted. Write $\mu_m = \sigma^{\ell_m - \ell_k}$, so that $y_m = \mu_m \eta_m$. The chain rule and the bound for $\langle y \rangle$ give

$$\left|\partial_\eta^\alpha H\right| \le C_\alpha \, \mu^\alpha \, H^{1-|\alpha|} = C_\alpha \, H \prod_{m \neq k} \Big(\frac{\mu_m}{H}\Big)^{\alpha_m}, \qquad \mu^\alpha = \prod_{m \neq k} \mu_m^{\alpha_m}.$$

Two simple facts control each factor μ_m / H:

- if m is good, then $\mu_m \le 1 \le H$, so $\mu_m / H \le 1$;
- if m is bad, then $H \ge \mu_m |\eta_m|$, so $\mu_m / H \le |\eta_m|^{-1}$.

In both cases $\mu_m / H \le v_m^{-1}$.

Lemma 2 (Weighted derivatives)**.** *Let $|\alpha| \ge 2$ with $\alpha_m = 0$ for $m \notin \mathcal{O} \cup \mathcal{Z}$, and choose $i, j \in \mathcal{O} \cup \mathcal{Z}$ (possibly $i = j$) such that $\alpha' = \alpha - \delta_i - \delta_j$ has nonnegative entries. Then for $t \ge 2$,*

$$\left|\partial_\eta^\alpha G(t,\eta)\right| \le C_\alpha \, \big(\bar\Lambda_i \bar\Lambda_j\big)^{1/2} \prod_{m \neq k} v_m(\eta)^{-\alpha'_m},$$

where $\bar\Lambda_m = \int_{1/t}^1 \sigma^{\kappa_m - 1} d\sigma$ is at most $1/\kappa_m$ for $m \in \mathcal{O}$ and equals $\log t$ for $m \in \mathcal{Z}$.

In words: two derivatives in oscillatory directions are free, a borderline one among them costs at most $(\log t)^{1/2}$, and each further derivative in a bad direction costs a factor $1/|\eta_m|$. Further derivatives in good directions cost nothing.

Proof. Since $|\alpha| = |\alpha'| + 2$, we can split the bound for $\partial_\eta^\alpha H$ as

$$\mu^\alpha H^{1-|\alpha|} = \frac{\mu_i \mu_j}{H} \prod_{m \neq k} \Big(\frac{\mu_m}{H}\Big)^{\alpha'_m} \le \frac{\mu_i \mu_j}{H} \prod_{m \neq k} v_m^{-\alpha'_m}.$$

The product on the right does not depend on σ. It remains to integrate the first factor against $\sigma^{\ell_k} d\sigma$. Since $\sigma^{\ell_k} \mu_m^2 = \sigma^{\kappa_m - 1}$, the bound $H \ge 1$ and Cauchy–Schwarz give

$$\int_{1/t}^1 \sigma^{\ell_k} \, \frac{\mu_i \mu_j}{H} \, d\sigma \le \Big(\int_{1/t}^1 \sigma^{\kappa_i - 1} d\sigma \Big)^{1/2} \Big(\int_{1/t}^1 \sigma^{\kappa_j - 1} d\sigma \Big)^{1/2} = \big(\bar\Lambda_i \bar\Lambda_j\big)^{1/2}.$$

Finally $\bar\Lambda_m = (1 - t^{-\kappa_m})/\kappa_m \le 1/\kappa_m$ if $\kappa_m > 0$, and $\bar\Lambda_m = \log t$ if $\kappa_m = 0$. □

4.4 What the weights suggest

Lemma 2 is made for the scale of stationary phase. At the scale $\lambda^{-1/2}$, a further derivative in a bad direction m costs $\lambda^{-1/2}/|\eta_m|$, which is at most 1 as soon as $|\eta_m| \ge \lambda^{-1/2}$. So the weighted bound controls all derivatives except in thin strips around the hyperplanes $\eta_m = 0$. A strip has width about $\lambda^{-1/2}$, exactly the gain oscillation would give, so it can be integrated trivially. Section 5 carries this out.

5 The stationary phase theorem at the zoom scale

5.1 Smooth sectors

The stationary phase estimate below needs a smooth amplitude, and the indicator of a sector is not smooth. So we replace the sharp sectors by a smooth partition. Fix smooth $\rho_1, \dots, \rho_d$ on $\mathbb{S}^{d-1}$ with $\sum_k \rho_k = 1$ and $\rho_k(p) = 0$ unless $|p_k| \geq \frac{1}{2\sqrt{d}}$. Such a partition exists since $\max_k |p_k| \geq 1/\sqrt{d}$ on $\mathbb{S}^{d-1}$. We insert $\rho_k(A_t\xi/|A_t\xi|)$ into I_k in place of the indicator of sector k. The η integral then runs over $\mathbb{R}^{d-1}$, with amplitude

$$b(\xi_k, \eta) = (1+|\eta|^2)^{-1/2}\, \tilde{\chi}(\xi_k, \eta)\, \rho_k\Big(\frac{\xi_k}{|\xi_k|}\, \frac{e_k + \sum_{j\neq k} \eta_j e_j}{(1+|\eta|^2)^{1/2}}\Big),$$

where $\tilde{\chi}(\xi_k, \eta) = \chi\big(\xi_k e_k + \sum_{j\neq k} t^{\ell_k - \ell_j} \xi_k \eta_j\, e_j\big)$ is the cutoff in the new variables. The kth component of the argument of ρ_k is $\pm(1+|\eta|^2)^{-1/2}$, so $b(\xi_k, \cdot)$ is smooth and supported in $|\eta| \leq 2\sqrt{d}$. The bounds of Section 2.4 on the support remain true with other constants: $|\xi_k| \leq 2$ and $\lambda \geq c\, t^{1+\ell_{\min}}$. Moreover $|\xi_m| \leq 2$ for every m on the support of $\tilde{\chi}$, that is,

$$|\eta_m| \leq \frac{2\, t^{\ell_m - \ell_k}}{|\xi_k|} \qquad (m \neq k). \tag{1}$$

5.2 The theorem

For $b \in C_0^\infty(\mathbb{R}^n)$ with support K, let K_1 be the set of points at sup distance at most 1 from K, and set

$$M_m = \sum_{2\leq|\alpha|\leq m} \sup_{K_1} |\partial^\alpha \Phi|, \qquad N_m = \sum_{|\alpha|\leq m} \sup_{K} |\partial^\alpha b|, \qquad a_0 = \inf_{K_1} \big|\det \nabla^2 \Phi\big|.$$

Theorem 3 (Alazard, Burq, Zuily, Theorem 3)**.** *Let Φ be real and smooth near K_1, with $a_0 > 0$ and $\nabla\Phi$ injective on K_1. Then for all $\lambda \geq 1$,*

$$\Big| \int_{\mathbb{R}^n} e^{i\lambda\Phi(\eta)}\, b(\eta)\, d\eta \Big| \leq \frac{C_n}{a_0} \big(1 + M_{n+2}\big)^{n/2}\, N_{n+1}\, \lambda^{-n/2},$$

where C_n depends only on n.

Two features of this bound matter for us. It involves no volume of K, so a large support costs nothing. And it sees Φ only through derivatives of order at least two, so the linear part of Φ, which carries x, plays no role.

5.3 Why the unit scale is not enough

Applied directly in η, Theorem 3 has its hypotheses satisfied for every fixed t: $a_0 \geq c$ by Lemma 1, and $\nabla\Phi$ is injective by concavity. But its constant is not uniform in t. Each ∂_{η_j} falling on $\tilde{\chi}$ produces the factor $t^{\ell_k - \ell_j}\xi_k$, so N_d can be as large as $t^{d(\ell_k - \ell_{\min})}$. By Section 4, the phase derivatives in M_{d+1} grow with t in bad directions. Borderline directions add a milder problem: Lemma 1 can give a_0 a factor $\log t$ for each of them, but their second derivatives in M are then of size $\log t$ as well, and the factor $(1+M)^{n/2}$ can cancel the gain.

All of this comes from measuring derivatives at unit scale in η, while the integral lives at scale $\lambda^{-1/2}$, and at scale $(\lambda \log t)^{-1/2}$ in borderline directions. We now check the hypotheses at that scale.

5.4 The zoom scale

Early and late switch. If there is a bad direction, we say that the switch is *early* if $|\xi_k| \geq 4t^{-\theta}$, and *late* otherwise. By (1) and the switch time of Section 3, an early switch means that the weight $s^{\ell_k}|\xi_k|$ overtakes every bad weight before the time $s = t^{1/2}$; this is made precise in part (a) of the proof below. If there is no bad direction, there is no borderline direction either, and the distinction plays no role.

Active candidates and scales. The directions used for stationary phase are

$$\mathcal{A} = \mathcal{O} \cup \mathcal{Z} \ \text{ (early switch)}, \qquad \mathcal{A} = \mathcal{O} \ \text{ (late switch)}.$$

Flat directions, and borderline directions after a late switch, are never used for stationary phase. The curvature scale of direction j is

$$L_j = \log t \ \text{ if } j \in \mathcal{Z} \text{ and the switch is early}, \qquad L_j = 1 \ \text{ otherwise}.$$

We take $t_0 \geq e$, so that $L_j \geq 1$.

The zoom. Put $\eta_j = (\lambda L_j)^{-1/2}\omega_j$ for every $j \neq k$. Then

$$J = \lambda^{-n/2} \prod_{j\neq k} L_j^{-1/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}(\omega)}\, \tilde{b}(\omega)\, d\omega, \qquad \tilde{\Phi}(\omega) = \lambda\, \Phi(\eta), \quad \tilde{b}(\omega) = b(\xi_k, \eta),$$

and by the chain rule

$$\partial_\omega^\alpha \tilde{\Phi} = \lambda^{1-|\alpha|/2} \prod_{m\neq k} L_m^{-\alpha_m/2}\, \partial_\eta^\alpha \Phi.$$

For $|\alpha| = 2$ this reads $\partial_{\omega_i}\partial_{\omega_j}\tilde{\Phi} = (L_iL_j)^{-1/2}\partial_{\eta_i}\partial_{\eta_j}\Phi$: the Hessian is normalized by the curvature scales. The decay $\lambda^{-n/2}$ and the gain $\prod_j L_j^{-1/2}$ now sit in front, and the ω integral only has to be bounded. This is why borderline directions are zoomed by $(\lambda \log t)^{1/2}$ rather than $\lambda^{1/2}$: the $\log t$ that Lemma 1 puts into a_0 moves into the Jacobian, where nothing can cancel it.

Strips. Let $\mathcal{B}_\mathcal{A}$ be the bad directions in $\mathcal{A}$: the bad oscillatory ones, and also the borderline ones after an early switch. Take $\vartheta \in C_0^\infty(\mathbb{R})$ with $\vartheta = 1$ on $[-2, 2]$ and $\vartheta = 0$ outside $[-3, 3]$. For each $m \in \mathcal{B}_\mathcal{A}$ write $1 = \vartheta(\omega_m) + (1 - \vartheta(\omega_m))$ and expand. This splits $J = \sum_S J_S$ over the subsets $S \subset \mathcal{B}_\mathcal{A}$ of directions placed in strips, with

$$J_S = \lambda^{-n/2} \prod_{j\neq k} L_j^{-1/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}}\, \tilde{b}_S\, d\omega, \qquad \tilde{b}_S = \tilde{b} \prod_{m\in S} \vartheta(\omega_m) \prod_{m\in\mathcal{B}_\mathcal{A}\setminus S} \big(1 - \vartheta(\omega_m)\big).$$

In J_S the *active* directions are $A = \mathcal{A} \setminus S$. All other directions, those in S and those outside $\mathcal{A}$, are *frozen*.

5.5 Checking the conditions at the zoom scale

Lemma 4 (Conditions at the zoom scale)**.** *Let $t \geq t_0$ and $S \subset \mathcal{B}_\mathcal{A}$, and fix the frozen variables. View $\tilde{\Phi}$ and $\tilde{b}_S$ as functions of $\omega_A \in \mathbb{R}^{|A|}$, and let $\tilde{K}$, $\tilde{K}_1$, a_0, $\tilde{M}$, $\tilde{N}$ be the quantities of Theorem 3 for them. Then $\nabla_{\omega_A}\tilde{\Phi}$ is injective, and*

$$a_0 \geq c, \qquad \tilde{M}_{|A|+2} \leq C, \qquad \tilde{N}_{|A|+1} \leq C,$$

with c, C depending only on d and the ℓ_j.

Proof. We first locate $\tilde{K}_1$ in the η variables. On $\tilde{K}$ the amplitude does not vanish, so $|\eta| \le 2\sqrt{d}$ and (1) holds. Passing to $\tilde{K}_1$ moves each active η_j by at most $(\lambda L_j)^{-1/2} \le 1$. For an active bad m this move is

$$(\lambda L_m)^{-1/2} = L_m^{-1/2}\, t^{-\kappa_m/2} |\xi_k|^{1/2} \cdot \frac{t^{\ell_m - \ell_k}}{|\xi_k|} \le \sqrt{2}\, \frac{t^{\ell_m - \ell_k}}{|\xi_k|},$$

because $\kappa_m \ge 0$ and $|\xi_k| \le 2$. Hence on $\tilde{K}_1$

$$|\eta| \le 3\sqrt{d}, \qquad |\eta_m| \le \frac{4\, t^{\ell_m - \ell_k}}{|\xi_k|} \quad (m \text{ bad}). \tag{2}$$

(a) The determinant a_0. By (2) and Lemma 1 with $R = 3\sqrt{d}$, $-\langle \nabla_\eta^2 \Phi\, v, v\rangle \ge c \sum_j \Gamma_j v_j^2$ on $\tilde{K}_1$. We claim $\Gamma_j \ge c\, L_j$ for $j \in A$. If $L_j = 1$ this is Lemma 1(i). If $L_j = \log t$, then j is borderline and the switch is early, and (2) gives for every bad m

$$|\eta_m|^{1/(\ell_k - \ell_m)} \le \Big(\frac{4}{|\xi_k|}\Big)^{1/(\ell_k - \ell_m)} t^{-1} \le t^{-1/2},$$

because $|\xi_k| \ge 4t^{-\theta} \ge 4t^{-(\ell_k - \ell_m)/2}$. So $\sigma_*(\eta) \le t^{-1/2}$, and Lemma 1(ii) gives $\Gamma_j \ge \frac{1}{2} \log t$. Now let $u \in \mathbb{R}^{|A|}$ and put $v_j = L_j^{-1/2} u_j$ for $j \in A$ and $v_j = 0$ otherwise. By the formula for the Hessian in ω,

$$-\langle \nabla^2_{\omega_A} \tilde{\Phi}\, u, u\rangle = -\langle \nabla^2_\eta \Phi\, v, v\rangle \ge c \sum_{j \in A} \frac{\Gamma_j}{L_j}\, u_j^2 \ge c\, |u|^2.$$

So every eigenvalue of $-\nabla^2_{\omega_A} \tilde{\Phi}$ is at least c, and $a_0 \ge c^{|A|}$. In the η variables this says $|\det \nabla^2_{\eta_A} \Phi| \ge c\, (\log t)^{|A \cap \mathcal{Z}|}$: the log in the determinant is exactly the gain $\prod_j L_j^{-1/2}$ in front of J_S.

(b) Injectivity. $\tilde{\Phi}$ is strictly concave in ω_A on all of $\mathbb{R}^{|A|}$, by Section 3 and the linear change of variables. So for $\omega \ne \omega'$,

$$\big\langle \nabla \tilde{\Phi}(\omega) - \nabla \tilde{\Phi}(\omega'), \omega - \omega' \big\rangle = \int_0^1 \big\langle \nabla^2 \tilde{\Phi}\big(\omega' + s(\omega - \omega')\big)(\omega - \omega'), \omega - \omega' \big\rangle\, ds < 0,$$

and $\nabla_{\omega_A} \tilde{\Phi}$ is injective.

(c) The phase derivatives $\tilde{M}$. Let $2 \le |\alpha| \le |A| + 2$ with α supported in $A \subset \mathcal{O} \cup \mathcal{Z}$, and choose i, j in the support of α. Then $\partial^\alpha \Phi = -\partial^\alpha G$, and the chain rule with Lemma 2 gives

$$|\partial_\omega^\alpha \tilde{\Phi}| \le C \Big(\frac{\bar{\Lambda}_i \bar{\Lambda}_j}{L_i L_j}\Big)^{1/2} \prod_{m \ne k} \big((\lambda L_m)^{1/2} v_m\big)^{-\alpha'_m}.$$

The first factor is bounded, since $\bar{\Lambda}_m \le C$ when $L_m = 1$ and $\bar{\Lambda}_m = \log t = L_m$ for active borderline m. In the product, a good m has $(\lambda L_m)^{1/2} v_m \ge \lambda^{1/2} \ge 1$. An active bad m lies in $\mathcal{B}_{\mathcal{A}} \setminus S$ and has $(\lambda L_m)^{1/2} v_m = |\omega_m|$. Since $1 - \vartheta(\omega_m)$ vanishes for $|\omega_m| \le 2$ and $\tilde{K}_1$ extends the support by sup distance 1, we have $|\omega_m| \ge 1$ on $\tilde{K}_1$. So every factor is at most 1, and $\tilde{M}_{|A|+2} \le C$.

(d) The amplitude derivatives $\tilde{N}$. One ω_j derivative, $j \in A$, is $(\lambda L_j)^{-1/2} \partial_{\eta_j}$. Falling on $(1 + |\eta|^2)^{-1/2}$ or on ρ_k, it costs at most $C(\lambda L_j)^{-1/2} \le C$. Falling on $\tilde{\chi}$, it produces the factor $t^{\ell_k - \ell_j} \xi_k$, so it costs

$$(\lambda L_j)^{-1/2}\, t^{\ell_k - \ell_j} |\xi_k| = L_j^{-1/2}\, t^{-\kappa_j/2}\, |\xi_k|^{1/2} \le \sqrt{2},$$

because $\kappa_j \ge 0$, $L_j \ge 1$ and $|\xi_k| \le 2$. This is where $\kappa_j \ge 0$ is used. Derivatives of the strip cutoffs are $O(1)$. So $\tilde{N}_{|A|+1} \le C$. □

The check in one table:

	unit scale in η	zoom scale in ω
a_0	$\geq c\,(\log t)^{\lvert A\cap\mathcal{Z}\rvert}$, offset by M	$\geq c$, gain moved to the Jacobian
M	grows with t in bad directions	$\leq C$ off the strips
N	up to $t^{d(\ell_k-\ell_{\min})}$	$\leq C$ in $\mathcal{O}\cup\mathcal{Z}$

6 Proof of the main estimate

6.1 Statement

Recall from Section 5 that $b(\xi_k,\cdot)$ is smooth and supported in $|\eta|\leq 2\sqrt{d}$, and that $|\xi_k|\leq 2$ and $\lambda\geq c\,t^{1+\ell_{\min}}$ whenever $b(\xi_k,\cdot)$ is not identically zero.

Theorem 5 (Main estimate)**.** *There are t_0 and C, depending only on d and $\ell_1,\dots,\ell_d$, such that for $t\geq t_0$, all $x\in\mathbb{R}^d$ and all $\xi_k\neq 0$,*

$$\Big|\int_{\mathbb{R}^{d-1}} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta\Big|\leq C\,\lambda^{-\frac{d-1}{2}}\prod_{j\in\mathcal{F}}t^{\kappa_j/2}\,|\xi_k|^{-1/2}\prod_{j\in\mathcal{Z}}E(\xi_k),\qquad \lambda=t^{\ell_k+1}|\xi_k|,$$

where $E(\xi_k)=(\log t)^{-1/2}$ if $|\xi_k|\geq 4t^{-\theta}$ and $E(\xi_k)=|\xi_k|^{-1/2}$ otherwise.

Each factor in the product over $\mathcal{F}$ is at most $|\xi_k|^{-1/2}$, since $\kappa_j<0$ there. When every direction is oscillatory, the estimate is the full stationary phase rate $\lambda^{-(d-1)/2}$. Each borderline direction gives an extra factor $(\log t)^{-1/2}$ after an early switch. After a late switch it is treated like a flat direction with $\kappa_j=0$.

Remark. The factors $|\xi_k|^{-1/2}$ have to be handled in the outer ξ_k integral of I_k. So does the late-switch region $|\xi_k|<4t^{-\theta}$, which has length at most $8t^{-\theta}$.

6.2 Proof

Write J for the integral in Theorem 5, with t, x, ξ_k fixed. If $b(\xi_k,\cdot)\equiv 0$ there is nothing to prove, so we assume $|\xi_k|\leq 2$ and $\lambda\geq c\,t^{1+\ell_{\min}}$, and we take $t_0\geq e$ so large that $\lambda\geq 1$.

Step 1: zoom and strips. By Section 5.4, $J=\sum_S J_S$ over the $2^{|\mathcal{B}_\mathcal{A}|}$ subsets $S\subset\mathcal{B}_\mathcal{A}$, with the decay $\lambda^{-n/2}\prod_j L_j^{-1/2}$ in front of each J_S.

Step 2: stationary phase in the active variables. Fix S and the frozen variables. By Lemma 4, Theorem 3 applies to the ω_A integral with $\lambda=1$ and gives

$$\Big|\int_{\mathbb{R}^{|A|}} e^{i\tilde{\Phi}}\,\tilde{b}_S\,d\omega_A\Big|\leq C,$$

uniformly in x and in the frozen variables.

Step 3: trivial integration in the frozen variables.

- $m\in S$. The cutoff $\vartheta(\omega_m)$ is supported in $|\omega_m|\leq 3$, so this costs a factor 6.

- $j \notin \mathcal{A}$, a flat direction or a borderline one after a late switch. Then $L_j = 1$, and (1) gives $|\omega_j| \le W_j$ with

$$W_j = \lambda^{1/2} \cdot \frac{2t^{\ell_j - \ell_k}}{|\xi_k|} = 2\, t^{\kappa_j/2}\, |\xi_k|^{-1/2} \le 2\, |\xi_k|^{-1/2},$$

so this costs a factor $2W_j$. Here $\kappa_j \le 0$ is used: the support in ω_j is no wider than a constant times $|\xi_k|^{-1/2}$.

Step 4: conclusion. Each piece satisfies

$$|J_S| \le C\, \lambda^{-\frac{d-1}{2}} \prod_{j \ne k} L_j^{-1/2}\, 6^{|S|} \prod_{j \notin \mathcal{A}} 2W_j.$$

After an early switch, $\prod_j L_j^{-1/2} = (\log t)^{-|\mathcal{Z}|/2}$ and the directions outside $\mathcal{A}$ are the flat ones. After a late switch, all $L_j = 1$ and the directions outside $\mathcal{A}$ are the flat and the borderline ones, with $t^{\kappa_j/2} = 1$ for the latter. Summing over S proves Theorem 5. The bound is uniform in x because x enters Φ only through its linear part, which none of a_0, $\tilde{M}$, $\tilde{N}$ sees.

Summary. After zooming to $\omega_j = (\lambda L_j)^{1/2}\eta_j$, the condition $\kappa_j \ge 0$ is exactly what keeps the amplitude smooth at scale 1, and Lemma 1 keeps the Hessian uniformly nondegenerate. Oscillatory directions gain $\lambda^{-1/2}$. Borderline directions gain $(\lambda \log t)^{-1/2}$ after an early switch, because every dyadic time scale after the switch adds the same amount of curvature. Flat directions have support thinner than the stationary phase scale and are integrated trivially. In bad directions the weighted bounds degenerate on a strip of unit width in ω, and integrating over the strip costs the same factor that oscillation would gain.